Adversarial Verification: Conditional Independence of Bootstrap-Trained Expert Error Indicators

Claim under review: If M experts are trained on independent bootstrap samples D_1, …, D_M (drawn i.i.d. with replacement from the dataset), then their error indicators e_m(x) = 1{ (f_m(x), y) > } are conditionally independent given the test point x.

Reference assumptions: The paper's A1 requires DISJOINT independent training sets (D_m D_{m'} =), which differs fundamentally from bootstrap. See below.

Attack 1: Bootstrap samples are NOT independent — the claim's premise is false

Description: The claim asserts "D_1, …, D_M are independent by construction (bootstrap)." This is mathematically incorrect.

Bootstrap with replacement from the same finite dataset D_original = {z_1, …, z_N} works as follows:

- D_m = {Z_{I_{m,1}}, …, Z_{I_{m,n}}} where each index I_{m,j} is drawn uniformly from {1, …, N} with replacement.
- D_1 and D_2 are functions of (D_original, _1) and (D_original, _2) respectively, where _1, _2 are independent index sets.

For a **fixed** D_original, D_1 and D_2 are conditionally independent given

D_original. But if D_original is itself random (which is the standard statistical setting — the training data is a random sample from population D), then:

P(D_1   A, D_2   B) = E[P(D_1   A | D_original) • P(D_2   B | D_original)]

Since P(D_1   A | D_original) and P(D_2   B | D_original) are both functions of the same random D_original, they are generally correlated. The expectation of the product does not factor:

E[U • V]   E[U] • E[V] when U, V are correlated.

Therefore, D_1 and D_2 are **not unconditionally independent**. Step 1 of the proof is wrong.

Concrete counterexample: Let D_original = {Z_1, Z_2} with Z_1, Z_2 ~ Bernoulli(p) i.i.d. (so N=2). Draw bootstrap samples of size n=1 (one element each). Then:

- P(D_1 = {1}) = P(Z_1=1 or Z_2=1) = 1 - (1-p)^2 = 2p - p^2
- P(D_1 = {1}, D_2 = {1}) = P(i,j: Z_i=1 and Z_j=1) = 1 - P(all Z's = 0) - 2 • P(exactly one Z=1, other draws that one) … more directly:

P(D_1={1}, D_2={1}) = P(at least one Z=1)^2 ? No. Let me compute:

P(D_1={1} | D_original) = 1{Z_1=1 or Z_2=1} P(D_2={1} | D_original) = 1{Z_1=1 or Z_2=1}

So P(D_1={1}, D_2={1}) = E[1{Z_1=1 or Z_2=1}^{2}] = P(Z_1=1 or Z_2=1) = 2p - p^2

And P(D_1={1}) • P(D_2={1}) = (2p - p^2)^2

These are equal only when $2p - p^2$ {0, 1}, i.e., p {0, 1}. For any non-degenerate p (0,1): $2p - p^2$ $(2p - p^2)^2$ because $2p - p^2$ (0,1).

Thus D_1 and D_2 are **not independent**. They are equal with positive probability due to both depending on the same D_original.

Verdict: FAIL — The very first step of the proof is false.

Attack 2: The proof sketch ignores the shared label y

Description: The proof states "e_m(x) depends only on f_m(x)." This is false. The error indicator is:

e_m(x, y) = 1{ (f_m(x), y) > }

It depends on **both** f_m(x) and the observed label y. Since y is a single random variable shared across all M experts, it creates a common source of randomness. Even if f_m(x) are independent, the indicators e_1, …, e_M all depend on the same y, so they are not independent unless y is degenerate.

Resolution: The claim can be salvaged if y is **independent** of the vector (f_1(x), …, f_M(x)). In the paper's setting, A4 states that the label noise is independent of all D_m and x. Under the paper's A1 (disjoint datasets), combined with a proper train-test split (x not in any D_m), we have:

- f_m(x) depends only on D_m
- y is independent of {D_1, …, D_M} (by train-test split + A4)
- Therefore, y is independent of (f_1(x), …, f_M(x))

In that case, e_m(x,y) = g(f_m(x), y) where g(a,b) = 1{ (a,b) > }. Since (f_1(x), …, f_M(x), y) are independent random variables, and e_m depends only on (f_m(x), y), the indicators e_1, …, e_M are **not** fully independent of each other (they share y), but they ARE conditionally independent given y (or given any -algebra that fixes y). However, the claim says "conditional independence given x" not "given x and y."

More precise analysis: Define e_m = 1{ (f_m(x), y) > }. Are e_1 and e_2 independent? Let's check:

P(e_1 = 1, e_2 = 1 | x) = E[P(e_1=1, e_2=1 | x, f_1, f_2, y) | x] = E[1{ (f_1(x), y) > } • 1{ (f_2(x), y) > } | x]

If y is independent of (f_1, f_2) given x, then: = P((f_1(x), y) > , (f_2(x), y) > | x, f_1, f_2) dP(f_1, f_2 | x)

The inner probability involves the same y, so it does not factor as P((f_1(x), y) > | f_1) • P((f_2(x), y) > | f_2). The indicators are **correlated through y**.

However, if we condition on y as well (i.e., "given x and y"), then:

P(e_1=1, e_2=1 | x, y) = P((f_1(x), y) > | x, y) • P((f_2(x), y) > | x, y)

which factors because f_1(x) and f_2(x) are independent given x, and y is now fixed by conditioning.

The proof's error: The proof says "e_m(x) depends only on f_m(x)" — this is simply wrong. It depends on f_m(x) AND y. The author appears to have forgotten that e_m takes two arguments, not one.

Verdict: FAIL — The proof sketch contains a concrete mathematical error. The claim IS true if we condition on both x and y, but the proof as written is incorrect.

Attack 3: Fatal mismatch between the claim and the paper's A1

Description: The paper's A1 states:

$$D_m \sim \mathcal{D}^{n_m}, \quad D_m \cap D_{m'} = \emptyset, \quad D_m \perp D_{m'} \text{ for } m \neq m'$$

This requires: 1. **Disjoint** training sets (no overlap) 2. Fresh i.i.d. draws from the population distribution

The claim substitutes "bootstrap samples drawn i.i.d. with replacement from the dataset," which is: 1. **Overlapping** (bootstrap resamples from the same data) 2. Drawn from a **single fixed dataset**, not from the population distribution

These are fundamentally different sampling regimes:

Property	Paper (A1)	Claim
Overlap?	Never (disjoint)	Frequent (with replacement)
Source	Population D	Fixed dataset
Independence	Unconditional	Conditional on dataset only
# of distinct samples	n_1 + … + n_M	At most N (original size)

The paper's Lemma 1, Lemma 2, and main theorem were constructed on the assumption of disjoint training sets. Swapping in bootstrap invalidates those proofs. For example, Lemma 1 computes:

E[C(x) | clean, x] = (1/M) Σ_m P((f_m(x), y) > | clean, x)

Under bootstrap, f_m are **not independent**, so the average of e_m has different concentration properties than what the paper's Chernoff bounds assume.

Verdict: FAIL — The claim investigates a different setting than the paper's. The claim's premise (bootstrap) contradicts the paper's A1 (disjoint).

Attack 4: Non-deterministic training and hidden shared randomness

Description: The claim assumes f_m = A(D_m) where A is deterministic. In practice:

- Deep learning uses stochastic gradient descent with random minibatch order
- Random initialization
- Data augmentation randomness (random crops, flips)
- GPU non-determinism (atomic operation ordering)
- Dropout

If A is not deterministic, then f_m = A(D_m, _m) where _m captures the training noise. The claim's step 2 ("f_m depends only on D_m") becomes false.

However: This is **fixable** if we assume the _m are independent across experts and independent of D_m. Then the argument extends: f_m depends on (D_m, _m), all of which are independent across m. The claim would then hold conditional on x and y (assuming the _m and D_m are independent of (x, y)).

Edge case: If all experts use the same random seed, _m =   for all m, creating shared randomness across experts. In this case, f_1, …, f_M share the same , breaking conditional independence. This is a real practical concern.

Edge case: GPU non-determinism is not "randomness" in the usual sense — it's deterministic chaos from parallel floating-point operations. It cannot be modeled as independent _m without explicit intervention (e.g., forcing separate CUDA streams).

Verdict: PASS (with caveats) — In theory, the randomness can be made explicit and independent. In practice, shared seeds or GPU non-determinism can create subtle dependencies. This is not fatal to the mathematical claim if the independence of _m is explicitly assumed.

Attack 5: "Independent random functions" does not automatically imply pointwise independence for random test points

Description: The claim says "f_m are independent random functions, therefore f_m(x) are independent for fixed x." This is mathematically correct for a **deterministic** x. However:

- The paper's main results involve expectations over the test distribution, not a single fixed x.

- If x is random (X), the statement "e_1(X), …, e_M(X) are independent given X" is a statement about regular conditional probabilities.
- For continuous X, the event {X = x} has measure zero, requiring careful use of regular conditional probabilities.

This is technically standard — the almost-sure definition of conditional independence works. So this is not a genuine flaw, merely a technical subtlety.

Verdict: PASS — Standard measure-theoretic conditioning handles this.

Attack 6: In the bootstrap setting, overlapping training data creates dependence beyond D_original

Description: Even when D_original is treated as fixed (conditional inference), bootstrap samples are conditionally independent given D_original. However, there is a deeper issue:

Two bootstrap samples D_1 and D_2 can share specific training examples. For instance, if D_original contains a unique outlier z* (an image with a rare artifact), and both D_1 and D_2 include z*, both f_1 and f_2 will have seen this outlier. Their predictions on test points similar to z* will be correlated in a way that is NOT captured by the independence assumption — they share information about z*.

Formally: P(f_1(x) = c | D_original) and P(f_2(x) = c | D_original) are both influenced by the same D_original. The conditional independence given D_original is fine, but the unconditional distribution of f_1 and f_2 (integrating over D_original) reflects the fact that both experts "see" the same data generating process.

For the paper's concentration bounds, what matters is:

$P(|C(x) - E[C(x)]| > \ \ | x)$

If we use the conditional independence given D_original, we get:

$P(|C(x) - E[C(x)]| > \ \ | x, D\_original) \ \ 2 \exp(-2M \ ^2)$

But then taking expectations over D_original:

$P(|C(x) - E[C(x)]| > \ \ | x) \ \ E[2 \exp(-2M \ ^2) | x] = 2 \exp(-2M \ ^2)$

Wait — this works because the bound is constant! In this case, the unconditional bound is the same. But if the bound depends on D_original (e.g., if the mean   varies with D_original), then:

$P(|C(x) - E[C(x)]| > \ \ | x) = E[P(|C(x) - E[C(x)]| > \ \ | x, D\_original)] \ \ E[2 \exp(-2M \ ^2)] = 2 \exp(-2M \ ^2)$ ← still OK because bound is constant

Actually, the issue is more subtle for the Chernoff bound used in the paper. The bound depends on the mean _s, which is an unconditional expectation. If we only have conditional independence given D_original, the Chernoff bound would involve the conditional mean given D_original, which is random.

This attack is a variant of Attack 1 and is ultimately fatal to the bootstrap interpretation.

Verdict: FAIL (same root cause as Attack 1)

Attack 7: The label y might come from the same distribution as the training data, creating subtle dependence

Description: In the paper's label noise model (A4), the observed label y is:

y = y* (true label) with probability 1- y = Uniform(Y {y*}) with probability

The label noise is independent of D_m. But the true label y* = f*(x) depends on x, which is the test point. If x is from the test set (properly separated), then y* is independent of D_m, and the claim holds.

But what if the claim's "bootstrap" setting includes the possibility that x comes from the training data? In bootstrap validation (like out-of-bag estimation), some test points may have been seen in some bootstrap samples. If x D_1 (overlap between train and "test"), then:

- f_1(x) is trained on x itself, creating a data leakage
- y is not independent of D_1 (since the pair (x, y) might be in D_1)
- e_1(x) and e_2(x) now have a complex dependence structure through the shared (x, y) in D_1

This breaks the conditional independence argument entirely. For a proper test point (not in any training set), this doesn't apply. But the claim doesn't specify train-test separation.

Verdict: FAIL — The claim uses "test point x" language but doesn't ensure x is not in any bootstrap sample. For proper evaluation (new unseen test data), this can be assumed away, but the bootstrap framing makes it ambiguous.

Final Verdict

The claim IS NOT rigorous. There are 4 significant issues, 3 of which are fatal:

#	Attack	Severity	Verdict
1	Bootstrap samples are not independent	**Fatal** — step 1 of proof is false	FAIL
2	Proof ignores the shared label y	**Fatal** — proof contains mathematical error	FAIL
3	Mismatch with paper's A1 (disjoint vs bootstrap)	**Fatal** — different sampling regime	FAIL
6	Bootstrap overlap creates dependence through D_original	**Fatal** — variant of Attack 1	FAIL
7	Training-test leakage in bootstrap	**Fatal** — not specified	FAIL
4	Non-deterministic training	Caveat, fixable	PASS
5	Formal conditioning issues	Standard technicality	PASS

Bottom line: The claim suffers from a fatal contradiction in its premises ("bootstrap" and "independent" are incompatible), a concrete mathematical error in its proof (ignoring the shared label y), and a mismatch with the paper's A1 assumption. Under the paper's actual assumption (disjoint datasets), and with a corrected proof that properly accounts for y, the conditional independence claim can be made rigorous — but the claim as stated and its proof sketch are not correct.